

PRATHIK ANAND KRISHNAN

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CAREER SUMMARY

Low-latency systems engineer with 3+ years of experience building performance-critical applications including trading and market-data infrastructure in C++.

Experienced in developing order execution management systems (OEMS), real-time market data pipelines, and concurrency-sensitive components for Crypto HFTs.

Currently pursuing an MSc in Software Engineering at the University of Amsterdam (expected Aug 2026).

SKILLS & TECHNOLOGIES

Technical Skills & Competencies	Technologies & Tools
- Web & Backend Microservice Development	- C++11,17,20, Python, Rust (learning), TypeScript
- Systems Engineering	- Multithreading, Asynchronous I/O, Event-Driven Architecture, STL, Boost.Asio, Boost.beast, Drogon
- Performance Engineering	- Memory layout, latency optimization, thread safety, profiling
- Trading Systems	- Centralized & Decentralized OEMS, Market-Data Pipelines, Blockchain AMM simulations
- Networking & Protocols	- REST, WebSockets, Protobufs, FlatBuffers, JSON-RPC
- Build & Integration toolchain, CI/CD	- GCC, MSVC, Meson, Conan, Azure pipelines
- Operating Systems familiarity	- Linux, Windows
- Testing, Automation & Profiling	- GoogleTests, Catch2, PyTest, Python Scripts, TDD, Vtune, Valgrind, perf
- Databases	- PostgreSQL, Redis, MySQL
- Agile delivery and Scrum processes	- Git, JIRA, MIRO board, Figma

WORK EXPERIENCE

BACKEND ENGINEER, Full-time – Remote

GoQuant – Miami, FL, USA

10/2024 – 12/2025

- Designed and implemented a **low-latency OEMS in C++17/Python**, integrating REST and WebSocket endpoints for order execution and market data pipelines for 4 exchange gateways (Bitmex, Bitfinex, Derebit, OKX) with support for **10+ order types** along with algorithmic strategies (TWAP, VWAP).
- Built asynchronous, **event-driven order monitoring system** for IOC, OCO, GTT, and DAY orders with thread-safe state updates.
- Collaborated in developing real-time **compliance, risk and execution-quality dashboards** using Redis, PostgreSQL, and FastAPI, with REST APIs for metrics such as PnL, latency, fill rates and slippage.
- Led a team of 3 members to develop **4 DEX adapters** (Uniswap v4, Raydium, Orca and Hyperliquid) enabling real-time trade execution, position tracking, and AMM simulations. Used optimized transaction construction to reduce avg block confirmation time by ~40% (from 800ms to 480ms).
- Built off-chain **AMM and Blockchain smart contract** interaction layer for Uniswap v4 and Orca DEX, supporting on-chain swaps via off-chain transaction signing.
- Developed **C++ WebSocket server** for multi-client symbol subscriptions with real-time order book streaming.
- Utilized Flatbuffers for high-performance market-data serialization, reducing overall system latency.
- Developed automated builds and testing scripts using Meson, Conan, and Shell scripts achieving end-to-end CI/CD integration and pipeline automation.

SOFTWARE DEVELOPER – G3

04/2023 – 10/2024

ARUP – Hyderabad, India

- Developed a C++20 static library Adapter API using the **Builder Pattern**, deployed as a NuGet package for seamless integration between proprietary applications.
- Improved UI responsiveness by offloading resource-heavy operations to a worker thread, with mutex-protected shared state to ensure data consistency and thread safety.
- Implemented **MFC UI** features and backend logics to effectively execute synchronization initiative
- Applied Test-Driven Development (TDD) using Google Test frameworks, achieving 96% code coverage in SDK and integrated SonarQube and CodeCov in CI/CD pipeline for continuous quality checks.
- Wrote Shell scripts for **Azure Pipelines** to automate build, deployment, and testing workflows.

SOFTWARE DEVELOPER – G2

09/2022 – 04/2023

ARUP – Hyderabad, India

- Created UML diagrams on **MIRO boards** for low-level and high-level system architecture visualization.
- Participated in pair programming and Tech-spikes to address technical debt.
- Built a PostHog-based event logging system via REST API, enabling real-time analytics and feature flag management within the application.
- I was also hosting Monthly Software Development Meets (OSM) at ARUP, creating a platform for knowledge sharing and was contributing to our team's technical growth.

Relevant Internships

- MATLAB developer | CSIR – National Geophysical Research institute, Hyderabad | June 2019 – Aug 2019

EDUCATION

- Masters in Software Engineering – University of Amsterdam | 2025 - 2026 | 8.2 / 10.0
- Masters in Structural Engineering – BITS Pilani, India | 2018 - 2020 | 9.04 / 10.0

Notable Projects

- [Order Execution System to trade on Deribit using C++17](#): Built an OMS with ~3.75 ms internal order processing latency, supporting order execution, market data, and real-time subscriptions.
- [VanillaVision - Twin Pricing Engine \(C++17\)](#): Built a multi-threaded option pricing engine comparing theoretical and simulation methods, achieving a 3x speedup over a single-threaded baseline while analyzing runtime-accuracy trade-offs on 10,000 option prices.
- [Black-Scholes Option Pricing Model Web App \(Python\)](#): This project provides an interactive Black-Scholes Option Pricing Model Web App that visualizes Call and Put option prices on a dynamic heatmap.
- [Other personal projects in C++17 and related web applications](#)

Notable Certifications & Accolades

- [MERIT 2022 winner](#) (Global Competition conducted by ICE – UK)
- Rated Top 36% in LeetCode.
- Certifications in [Mastering Advanced C++17](#), [Harvard CS50](#), [Meta full-stack development with JavaScript and React](#) and [Duke University – Java Programming](#) in Coursera.
- Scholarships: BITSAT Graduate Application Scholarship and PG scholarship from MHRD